REPUBLIC OF THE PHILIPPINES
CAGAYAN STATE UNIVERSITY
www.csu.edu.ph

.College of Information and Computing Sciences

UNIVERSITY PHILOSOPHY
The University serves the INDIVIDUAL by providing the student with a nurturing environment for optimal human flourishing. It serves the community by offering programs responsive to individual and societal needs.

VISION
CSU is a University with global stature in the arts, culture, agriculture and fisheries, the sciences as well as technological and professional fields.

MISSION
Cagayan State University shall produce globally competent graduates through excellent instruction, innovative and creative research, responsive public service and productive industry and community engagement.



SYLLABUS IN COMPUTER PROGRAMMING 2

Course Credits	**3**	**Semester**	**Second**	**School Year**	**2024-2025**	**Number of Lecture Hours**	**36**
Pre-requisite/s	**IT112**		**Co-requisite/s**	**None**		**Number of Laboratory Hours**	**54**

Course Description:	**This course is a continuation of IT112- Computer Programming 1. The emphasis is to train students to design, implement, test and debug programs intended to solve computing problems using basic data structures and standard libraries.**
Intended Graduate Attributes (IGAs)	
Course Intended Learning Outcomes(CILOs)	**CILO 1** Design, implement, test and debug a program, based on a given specification that uses (1) data structures arrays, strings, structures, linked lists and files, (2) conditional, iterative and recursive constructs, and (3) standard libraries in the assigned programming language.(IT01,IT02,IT05, IT07, IT09,IT10) **CILO 2** Assess and recommends revisions to another programmer's code (1) regarding appropriateness of chosen data structure, (2) regarding appropriateness of chosen conditional and iterative constructs given a programming task, and (3) regarding thoroughness in applying procedural abstraction. (IT01,IT02,IT05, IT07, IT09,IT10) **CILO 3** Argue the cost and benefits of dynamic and static data structure implementations. (IT01,IT02,IT05, IT07, IT09,IT10) **CILO on Internationalization** **CILO on Gender and Development, and other SGDs**
Grading System	Percentage Distribution Assigned Activity 15% Engagement/Forum 10% Quizzes/Unit Test 30% Major Examination 45% TOTAL 100% Laboratory: **NOTE:** If course is with Laboratory, the grade will be computed as **3 Units**: Lecture: 2 units (67%) Laboratory: 1 unit (33%) Total: 100% **4 Units**: Lecture: 3 units (75%) Laboratory: 1 unit (25%) Total: 100%

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	Laboratory Reports % Performance % Practical Exam/ % Midterm/Final Exam	**5 Units:** Lecture: 3 units (60%) Laboratory: 2 unit (40%) Total: 100%

Time Frame	Learning Content	Intended Learning Outcomes (ILOs)	Teaching and Learning Activities (TLAs)	Learning Assessments	Teaching and Learning References/ Resources	Modality
Week1	**1. Statement of the Vision, Mission, Goal and Objectives** a. University Vision b. University Mission c. College ILO d. Program ILO e. University Officials f. Student Policies	At the end of the session, students are expected to: ● Live by the importance of the University's and College's Vision, Mission, and Philosophy in the students' life in the University and College. ● Familiarized with the key officials of the University and College. ● Accord proper respect and line of communication for the concerns of the students.	Discussion Audio-Visual Presentation of University Officials	Recitation	University Manual Student Handbook University Website:https://csu.edu.ph/	Blended Learning

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Week 2 - Week 3	**User Defined Functions** 2.1 Function Syntax 2.2 Calling a Function 2.3 Function Scope 2.4 Variable Parameter List	At the end of the session, students are expected to: ● Create a function and declare local variables in functions(IT01, IT02, IT07,IT09, IT10) ● Develop programs using functions(IT05,IT07, IT09, IT10)	Uploading/ downloading of Modules using CSU LENA and other online platforms Online Discussions Discussion/Demonstration on the C# User-Defined Functions Library Work for the user-defined functions, its syntax and scope Lab Exercise on creating a user-defined function	Quiz 1 (20 items) Problem Drill 1 (15 items) Presentation of Output 1 (30 pts) Hands-Activity 1(30 pts)	https://dotnettutorials.net/lesson/user-defined-functions-in-csharp/	Blended Learning
Week 4 - Week 5	**3. The C# Array List** 3.1 What is Array List? 3.2 Creating an Array List 3.3 ArrayList Class 3.4 Array versus Array List 3.5 Operations on Array List (Adding and	At the end of the session, students are expected to: ● Use Array List to Store and retrieve data(IT01,IT02, IT05) ● Declare and Initialize ArrayList(IT01, IT05,IT07,IT09)	Uploading/ downloading of Modules using CSU LENS and other online platforms Online Discussions Discussion/demonstration on C# Array Lists Library Work for the	Quiz 2 (35 items) Problem Drill 2 (15 items) Presentation of Output 2 (30 pts) Hands-on	https://www.programiz.com/csharp-programming/arraylist https://www.tutorialsteacher.com/csharp/csharp-arraylist	Blended Learning

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	Removing Elements) 3.6 Working with Range Elements 3.7 Conversion of ArrayList to Array	• Create programs using arraylists • (IT01, IT05, IT07,IT09,IT10)	C# Array List, its syntax and scope Lab Exercise on creating a C# Array List	activity 3 (30 pts) Hands-on activity 4 (30 pts)		
Week 6	PRELIM EXAMINATION					
Week 7 - Week 8	**4. Strings and Lists** 4.1 String Class 4.2 String Properties and Methods 4.3 Verbatim String Literal 4.4 String Builder 4.5 String versus String Builder 4.6 List Properties and Methods	At the end of the session, students are expected to: • Create string classes, properties and methods(IT01, IT02, IT05,IT07,IT09) • Create and manipulate character string objects using class String and class StringBuilder(IT01, IT02, IT05,IT07,IT09) • Use various methods of classes String and StringBuilder(IT01, IT02, IT05,IT07,IT09) • Create a	Uploading/ downloading of Modules using CSU LENS and other online platforms Online Discussions Discussion/Demonstration on the C# Strings and Lists Discussion/Demonstration on String and Lists properties and methods Lab Exercise in using properties and methods on Strings and Lists	Quiz 3 (40 items) Problem Drill 3 (15 items) Presentation of Output 3 (30 pts) Hands-on activity 5 (30 pts) Problem Drill 4 (15 items) Quiz 4 (30 items)	https://www.programiz.com/csharp-programming/list https://www.c-sharpcorner.com/article/c-sharp-list/ https://www.tutorialsteacher.com/csharp/csharp-list https://www.techotopia.com/index.php/Working_with_Strings_in_C_Sharp	Blended Learning

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		List Collection (IT01, IT05,IT07 ,IT09) ● Manipulate programs with lists and methods (IT01, IT05,IT07,IT09) ● Develop programs using Strings and List(IT01, IT05,IT07,IT09,IT10)				
Week 9	**5. Structures** 5.1 Structure Within a Structure 5.2 Array Within a Structure	At the end of the session, students are expected to: ● Define what a structure is(IT01, IT02, IT10) ● Differentiate a structure from an array(IT01, IT02, IT10) ● Declare a structure type(IT01, IT02) ● Declare variables	Uploading/ downloading of Modules using CSU LENS and other online platforms Online Discussions Discussion/Demonstration on the C# Structures Lab Exercise on creating a C# Structures	Quiz 5 (25 items) Problem Drill 5 (15 items) Presentation of Output 4 (30 pts) Hands-on Activity 6 (30 pts)	https://www.tutorialspoint.com/csharp/csharp_struct.htm https://www.programiz.com/csharp-programming/struct	Blended Learning

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		as structures(IT01, IT10) ● Initialize structure members(I T01,IT05,I T07,IT09,I T10) ● Access structure members(I T01,IT05,I T07,IT09,I T10)				
Week 10- Week 11	**6. Hashtable s and Dictionaries** 6.1 Hashtables Properties and Methods 6.2 Dictionaries Properties and Methods	● Describe the purpose and use of a hashtable(IT01,IT0 2,IT10) ● Describe the purpose and use of dictionary(IT01,IT0 2,IT10) ● Describe the purpose and use of key/value pairs(IT01,IT02,IT 10) ● Describe the purpose of a key(IT01,IT02,IT1 0) ● List and describe the interfaces a class must	Uploading/ downloading of Modules using CSU LENS and other online platforms Online Discussions Discussion/Demon stration on the C# Hashtables and Dictionaries Discussion/Demon stration on the C# Hashtables and Dictionaries properties and methods Lab Exercise on creating	Quiz 6 (25 items) Problem Drill 6 (15 items) Presentation of Output 4 (30 pts) Hands-on activity 7 (30 pts) Quiz 7 (25 items) Problem Drill	https://www.geeksforgeeks.org/difference-between-hashtable-and-dictionary-in-c-sharp/ https://www.c-sharpcorner.com/blogs/difference-between-dictionary-and-hashtable-in-c-sharp Fun	Blended Learning

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		implement if it's to be used as a key(IT01,IT07,IT10) ● Describe the functionality provided by the IDictionary interface(IT01, IT02,IT10)	a C#Hashtables and Dictionaries	7 (15 items) Hands-on activity 8 (30 pts)		
Week 12	MIDTERM EXAMINATION					
Week 13-Week 14	7. **The C# Language-Integrated Query (LINQ)**	● Learn basic LINQ concepts(IT01, IT02, IT09) ● Querry an array using LINQ(IT01, IT05, IT07,IT09, IT10)	Uploading/ downloading of Modules using CSU LENS and other online platforms Online Discussions Discussion/Demonstration on the C# Language-Integrated Query Lab Exercise on creating a C# Language-Integrated Query	Quiz 7 (40 items) Problem Drill 8 (15 items) Presentation of Output 5 (30 pts) Hands-on activity 8 (30 pts)	https://www.tutorialsteacher.com/linq/what-is-linq https://www.geeksforgeeks.org/linq-language-integrated-query/ https://learn.microsoft.com/en-us/dotnet/csharp/linq/get-started/introduction-to-linq-queries	Blended Learning
Week 15-Week 17	**8. File Handling**	● Create, read, write, and update	Uploading/ downloading of Modules using CSU	Quiz 8 (40 items)	https://www.geeksforgeeks.org/basics-of-file-handling-in-c-sharp/	Blended Learning

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	8.1 Data Hierarchy 8.2 Files and Streams 8.3 Classes File and Directory 8.4 Text File	files(IT01,IT05, IT07, IT09) ● Use classes File and Directory to obtain information about files and directories(IT01, IT05,IT07,IT09,IT10) ● Use LINQ to search through directories(IT01, IT05,IT07,IT09,IT10) ● Use classes FileStream, StreamReader and StreamWriter to read objects from and write objects to files(IT01, IT05,IT07,IT09,IT10)	LENS and other online platforms Online Discussions Discussion/Demonstration on the C# File Handling Discussion/Demonstration on the C# files, streams and directory Lab Exercise on the C# file handling	Problem Drill 9 (15 items) Presentation of Output 6 (30 pts) Hands-on activity 9 (30 pts)	https://www.w3schools.com/cs/cs_files.php	
Week 18	FINAL EXAMINATION					

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INTENDED LABORATORY ACTIVITIES (if applicable)

1. Create a program using User-Defined functions
2. Create a program using ArrayList
3. Create a program using properties and methods on Strings and Lists
4. Create a program using C# Structures
5. Create a program using C#Hashtables and Dictionaries
3. Create a program using C# Language- Integrated Query(LINQ)
4. Create a program using C# file handling

OTHER MAJOR COMPETENCIES OUTPUT (i.e., Reflection Paper/Portfolio/Film Review, etc. *if applicable*)
Aside from the major examinations, the student will be assessed at other times during the term by the flowing:

- **Compilation of C# programs**

RUBRIC FOR ASSESSMENT TO BE USED FOR Major Competencies Outputs (MCOs) (if applicable)

Criteria	Excellent 9-10 pts.	Good 8-9 pts.	Satisfactory 5-6 pts.	Unsatisfactory Below 5 pts.
Functionality	The application fulfills all specified requirements without errors.	The application mostly fulfills requirements with minor bugs.	The application partially fulfills requirements with significant bugs.	The application does not meet most requirements.
Code Quality and Structure	Code is well-structured, easy to read, follows best practices.	Code is mostly well-structured but may lack consistency.	Code structure is disorganized and difficult to follow.	Code is poorly structured and lacks conventions/comments.
User Interface (UI) Design	UI is clean, intuitive, enhances user experience.	UI is functional but may lack polish or intuitive design.	UI is rudimentary, lacks essential features/aesthetic appeal.	UI is poorly designed, making navigation and interaction difficult.
Error Handling and Exception	Application gracefully handles	Error handling is present but	Error handling is minimal,	Application lacks error

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Management	errors, provides informative messages.	may lack detail/consistency.	application may crash unpredictably.	handling, frequently crashes unpredictably.

REFERENCES

Print

- **"C# Programming for Beginners: An Introduction to C# Programming and Building Console Applications"** by Troy Dimes (2014)
- **"Learn C# in One Day and Learn It Well: C# for Beginners with Hands-on Project (Learn Coding Fast with Hands-On Project)"** by Jamie Chan (2015)

Non-Print

- https://dotnettutorials.net/lesson/user-defined-functions-in-csharp/
- https://www.programiz.com/csharp-programming/arraylist
- https://www.tutorialsteacher.com/csharp/csharp-arraylist
- https://www.programiz.com/csharp-programming/list
- https://www.c-sharpcorner.com/article/c-sharp-list/
- https://www.tutorialsteacher.com/csharp/csharp-list
- https://www.techotopia.com/index.php/Working_with_Strings_in_C_Sharp
- https://www.tutorialspoint.com/csharp/csharp_struct.htm
- https://www.programiz.com/csharp-programming/struct
- https://www.geeksforgeeks.org/difference-between-hashtable-and-dictionary-in-c-sharp/
- https://www.c-sharpcorner.com/blogs/difference-between-dictionary-and-hashtable-in-c-sharp
- https://www.tutorialsteacher.com/linq/what-is-linq
- https://www.geeksforgeeks.org/linq-language-integrated-query/
- https://learn.microsoft.com/en-us/dotnet/csharp/linq/get-started/introduction-to-linq-queries
- https://www.geeksforgeeks.org/basics-of-file-handling-in-c-sharp/
- https://www.w3schools.com/cs/cs_files.php

Software

- **Microsoft Visual Studio 2013**
- **C# Shell**
- **SharpDevelop 5.1**

COURSE POLICIES

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(on classroom norms, i.e., attendance, missed exams, academic integrity and medium of communication (provided that they don't run counter to existing policies)

- Students are expected to attend classes regularly. If the student accumulates absences for at least 20% of the total number of class hours, the student shall be considered dropped. Dropping should be done before the midterm examination to be considered officially dropped, otherwise, considered failed (except in cases found meritorious).
- The medium of communication shall be English, except in those courses where the use of the national language is necessary.
- Students must adhere to the mandates of academic integrity. Academic dishonesty (cheating, plagiarism, etc.) is taken very seriously. The minimum penalty for a violation of academic integrity is a failure (zero) for the assignment. When using course materials or other credible sources during class discussions and written requirements these must be cited properly.
- Use of electronics during class is not allowed. Turn off/mute/set to vibrate any electronic devices before class begins. For personal emergency, the student asks to be excused from the class to communicate outside of the classroom.
- Missing an examination is not allowed. In extreme cases such as death in the family, sickness or accident, a written evidence is required to make up a missed exam. No make-up quizzes shall be allowed.
- Students are encouraged to contact their professor for consultation and feedbacking at the following contact information:

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